

TITLE:HR ANALYTICS DASHBOARD

Name:Jayshree Manohar Shinde

SYNTEXHUB INTERN

Employee Attrition Analysis using Power BI

1. Introduction

Human Resource Analytics helps organizations understand employee behavior, workforce trends, and attrition patterns. This project focuses on analyzing employee data using Power BI to identify key factors affecting employee attrition and organizational performance.

The dashboard was developed using Power BI and DAX to provide interactive visual insights for decision-making.

2. Objectives

The main objectives of this project are:

To analyze employee attrition patterns

To identify factors influencing employee turnover

To compare employee performance across departments

To perform correlation analysis between important variables

To create an interactive HR Analytics Dashboard

To provide business insights for improving employee retention

3. Tools & Technologies Used

Tool	Purpose
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Power bi	Dashboard Development
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DAX	KPI Measures & Calculation
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Excel/CSV Dataset	HR Data Source
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4. Dataset Description

The dataset contains employee-related information such as:

Employee ID

Age

Gender

Department

Job Role

Monthly Income

Years at Company

Attrition

Job Satisfaction

Education Field

Overtime

Performance Rating

The dataset was cleaned and transformed before visualization.

5. Data Cleaning & Preparation

The following preprocessing steps were performed:

Removed duplicate records

Checked missing values

Corrected data types

Converted categorical values into numeric format

Created calculated columns and DAX measures

Example:

Attrition converted:

Yes → 1

No → 0

6. KPI Measures Created

The following KPIs were developed:

Total Employees

Attrition Count

Attrition Rate

Retention Rate

7. Dashboard Visualizations

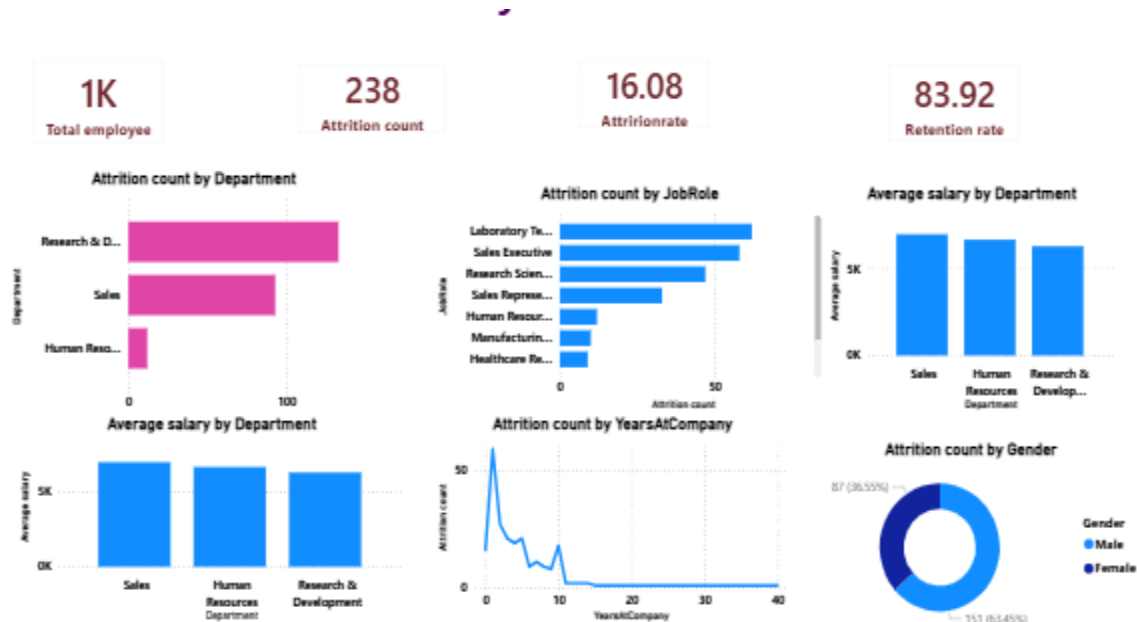
The dashboard includes the following visuals:

KPI Cards

Total Employees

Attrition Rate

CHARTS



1. Attrition Count by Department

Insight

The Research & Development department shows the highest employee attrition.

The Sales department also experiences considerable attrition.

The Human Resources department has the lowest attrition count.

Business Interpretation

High attrition in R&D and Sales may affect productivity and organizational performance.

HR management should focus on employee retention strategies in these departments.

2. Attrition Count by Job Role

Insight

Laboratory Technicians show the highest attrition among job roles.

Sales Executives and Research Scientists also experience higher turnover.

Manufacturing and Healthcare Representative roles have comparatively lower attrition.

Business Interpretation

Technical and sales-related roles may face higher workload or pressure.

Organizations may need better engagement and career development programs for these employees.

3. Average Salary by Department

Insight

The Sales department has the highest average salary.

Human Resources and Research & Development departments have slightly lower average salaries.

Business Interpretation

Salary differences across departments may influence employee satisfaction and retention.

Compensation policies should be reviewed to ensure fairness and motivation.

4. Attrition Count by Years at Company

Insight

Attrition is highest during the early years of employment.

Employee exits decrease significantly as years at company increase.

Business Interpretation

New employees are more likely to leave the organization.

The company should improve onboarding, training, and employee engagement for newly hired staff.

5. Attrition Count by Gender

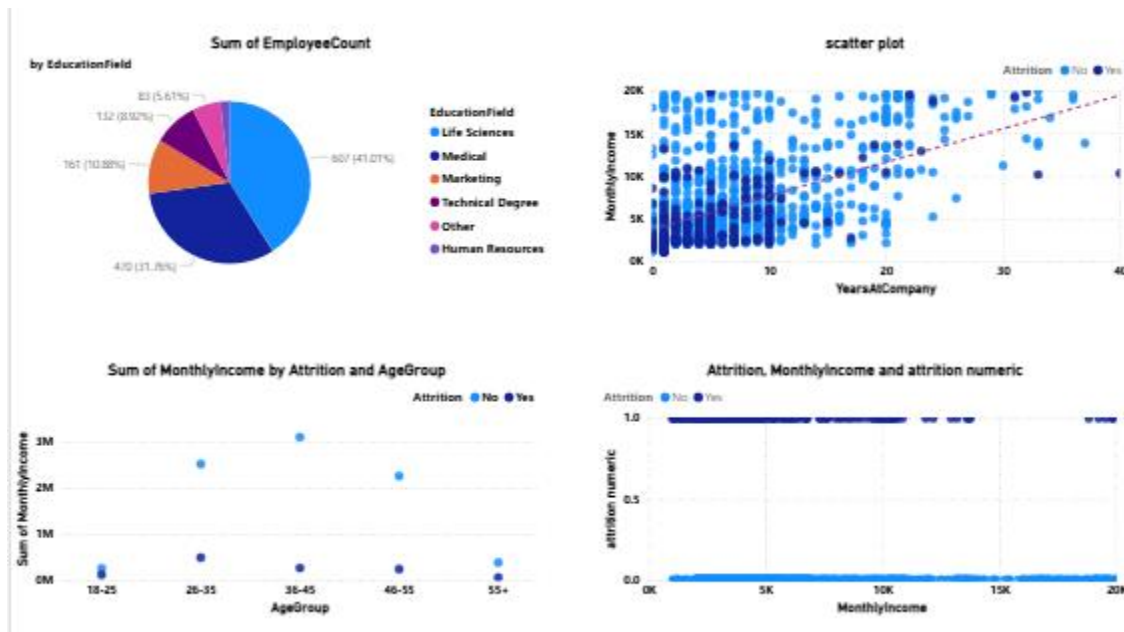
Insight

Male employees show slightly higher attrition compared to female employees.

Attrition exists across both genders but is more concentrated among male employees.

Business Interpretation

HR teams may analyze workload, work-life balance, and departmental distribution to understand gender-based attrition differences.



1. Sum of EmployeeCount by EducationField (Pie Chart)

The largest segment is light blue (Life Sciences), followed by dark blue (likely Engineering or Technical fields), with smaller portions in orange (Marketing), purple/magenta (possibly Technical Degree), and other categories. Life Sciences dominates the employee headcount.

2. Scatter Plot (Top Right)

Shows the relationship between Years at Company (x-axis) and what appears to be Monthly Income (y-axis). The dense clustering of blue dots suggests:

Most employees cluster in the lower-to-middle income range

Income generally increases with tenure, though there's significant variation

Some outliers earn substantially more despite similar tenure

The trend line shows a positive correlation between time at company and compensation

3. Sum of MonthlyIncome by Attrition and AgeGroup (Bottom Left)

Compares income levels across age groups (18-25, 26-35, 45-55, 55+), split by attrition status (Yes/No):

Income increases with age (older employees earn more)

The 55+ age group shows notably higher income

Attrition appears relatively consistent across age groups

4. Attrition, MonthlyIncome and Attrition Numeric (Bottom Right)

A trend analysis showing:

Attrition status (Yes/No) compared against monthly income

Both groups show similar income distributions with minimal difference

Suggests attrition is not strongly driven by salary alone

Key Insight: Employee retention may need attention beyond compensation, as attrition occurs across all income levels.

8.Key Insights

Total Employees: Approximately 1,000

Attrition Rate: 16.08%

Retention Rate: 83.92%

Research & Development department shows the highest attrition.

Laboratory Technicians and Sales Executives experience maximum employee turnover.

Employees with lower years at the company are more likely to leave.

Male employees show slightly higher attrition compared to female employees.

Salary increases with employee experience.

Early-stage employees require better engagement and retention strategies.

9.Business Recommendation

The organization should focus on:

Improving employee engagement programs

Strengthening onboarding processes

Providing career growth opportunities

Enhancing work-life balance

Monitoring high-attrition departments regularly

These strategies can help reduce employee turnover and improve long-term workforce stability.

10.Conclusion

The HR Analytics Dashboard provides valuable insights into employee attrition, workforce distribution, salary trends, and employee behavior within the organization. The analysis reveals that employee attrition is mainly concentrated in the Research & Development and Sales departments, particularly among Laboratory Technicians and Sales Executives.

The dashboard also indicates that employees with fewer years at the company are more likely to leave, highlighting the importance of effective onboarding, employee engagement, and retention strategies for newly hired employees. Salary and experience show a positive relationship, while lower satisfaction and workload-related factors may contribute to higher attrition.

Overall, the dashboard helps HR teams identify critical workforce challenges and supports data-driven decision-making to improve employee retention, employee satisfaction, and organizational performance.

